

## Circuit Analysis (Lab10)

### Experiment 9: Verification of Thevenin's Theorem

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Date: 6/May/2026 Grade and Signature: .....

|                       |                                                                                                                                                                                              |              |              |              |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|--------------|--------------|
| CLO2<br>CLO3          | Use with identification the circuit components, breadboards, multimeter, power supplies, signal generators, and oscilloscopes. Make measurements in their own constructed electric circuits. |              |              |              |
| Psychomotor/Affective | Level1<br>(1)                                                                                                                                                                                | Level2 (2-3) | Level3 (4-5) | Level4 (6-7) |
| Report Marks (3)      | Total marks<br>(10)                                                                                                                                                                          |              |              |              |

### Objective

The objective of this lab is verify Thevenin's theorem and to find the full load current.

### Apparatus Required

| Sr. no | Apparatus              | Range        | Quantity    |
|--------|------------------------|--------------|-------------|
| 1      | Regulated Power Supply | Standard     | 1           |
| 2      | Signal Generator       | Standard     | 1           |
| 3      | Oscilloscope           | Standard     | 1           |
| 4      | Digital Multimeter     | Standard     | 1           |
| 5      | Breadboard & Wires     | Standard     | As Required |
| 6      | Resistors              | 1k, 330 ohms | 1,3         |
| 7      | Potentiometer          | Standard     | 1           |

### Description

#### Statement:

Any linear bilateral, active two terminal network can be replaced by an equivalent voltage source (VTH). Thevenin's voltage or  $V_{OC}$  in series with looking back resistance  $R_{TH}$ .

**Precautions:**

1. Voltage control knob of RPS should be kept at minimum position.
2. Current control knob of RPS should be kept at maximum position

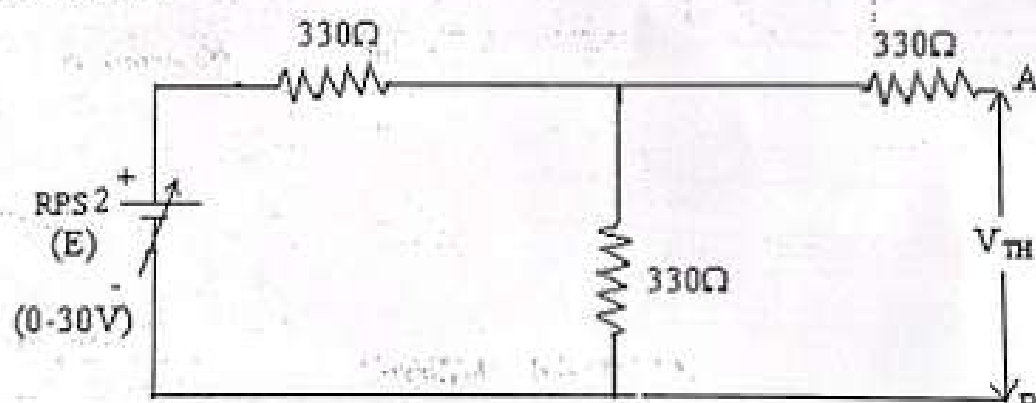
**Procedure:**

1. Connections are given as per the circuit diagram.
2. Set a particular value of voltage using RPS and note down the corresponding ammeter readings. To find  $V_{TH}$
3. Remove the load resistance and measure the open circuit voltage using multimeter ( $V_{TH}$ ). To find  $R_{TH}$
4. To find the Thevenin's resistance, remove the RPS and short circuit it and find the  $R_{TH}$  using multimeter.
5. Give the connections for equivalent circuit and set  $V_{TH}$  and  $R_{TH}$  and note the corresponding ammeter reading.
6. Verify Thevenin's theorem.

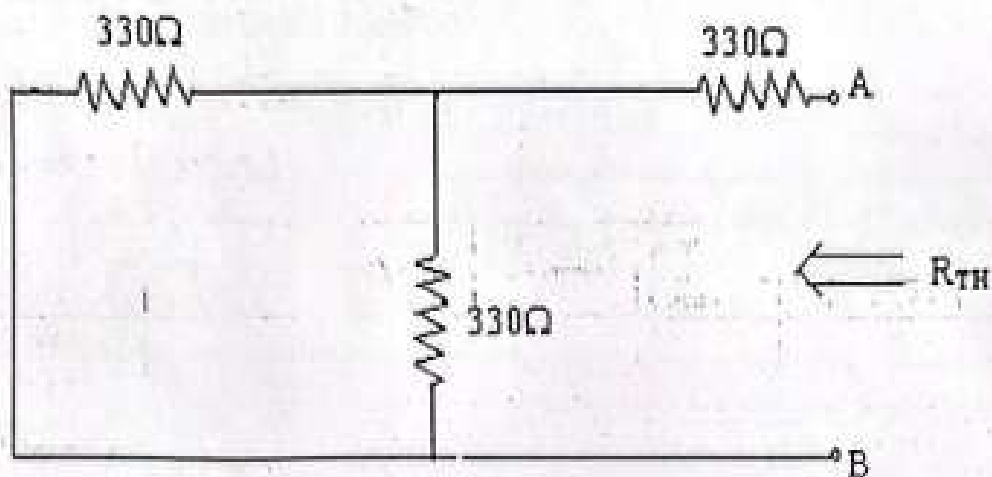
**Circuit - 1: To find load current**



**To find  $V_{TH}$**



To find  $R_{TH}$



Thevenin's Equivalent circuit:



Thevenin Practical Values

|        | E(V) | $V_{TH}(V)$ | $R_{TH}(\text{ohms})$ | $I$ (mA)    |                    |
|--------|------|-------------|-----------------------|-------------|--------------------|
|        |      |             |                       | Circuit - I | Equivalent Circuit |
| Values | 30 V | 15.0 V      | 495.00                | 10.0 mA     | 10.0 mA            |

## Post Lab Questions

### 1. What is Thevenin's Theorem?

#### Thevenin's Theorem:

Any linear 2-terminal electrical circuit can be replaced by an equivalent circuit consisting of 1) voltage source ( $V_{th}$ ) 2) A resistance ( $R_{th}$ ) in series with it.

This equivalent circuit makes complex circuit analysis easier.

### 2. What are the applications of Thevenin's Theorem?

- 1) It simplifies complex electrical circuits into a simple equivalent circuit.
- 2) It helps in finding voltage and current across a load resistor.
- 3) It is used in circuit and design.
- 4) It makes troubleshooting and testing of circuit easier.
- 5) It is useful in electronic and power system applications.

### 3. What are the theoretical and practical limitations of Thevenin's Theorem?

- 1) It applies only to linear electrical circuit.
- 2) It cannot be directly used for non-linear component like diodes and transistors.
- 3) It becomes difficult when circuits have many dependent sources.
- 4) In practical circuits, results may vary due to component tolerances and losses.
- 5) Finding Thevenin equivalent can be time consuming for very large networks.